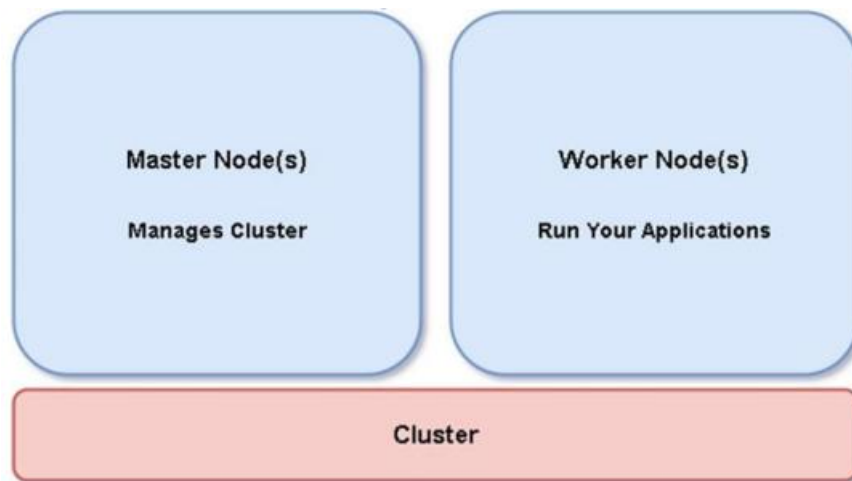


Kubernetes Cluster



1. Cluster:

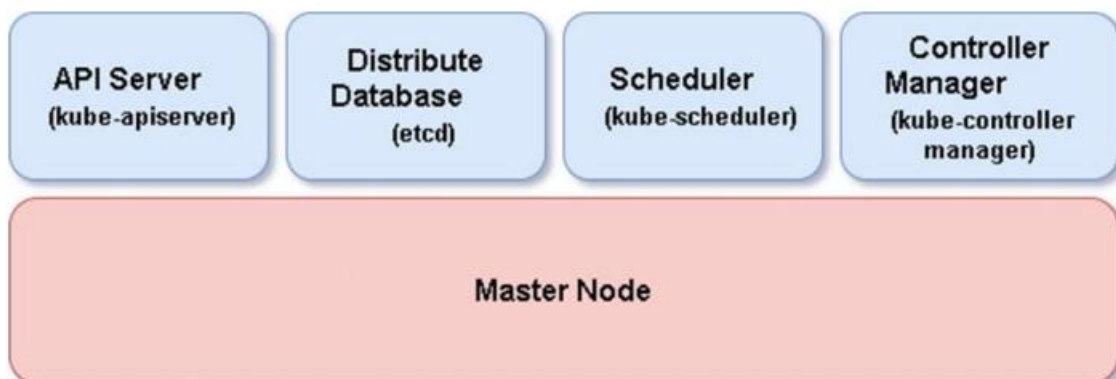
A combination of Master Node(s) and Worker Node(s).

2. Master Node:

Responsible for managing the cluster and making worker nodes available.

3. Worker Node:

Responsible for running the actual applications (Pods and containers).



1. API Server

- API server is used to carry the commands from kubectl (command prompt) to kubernetes cluster.

2. etcd

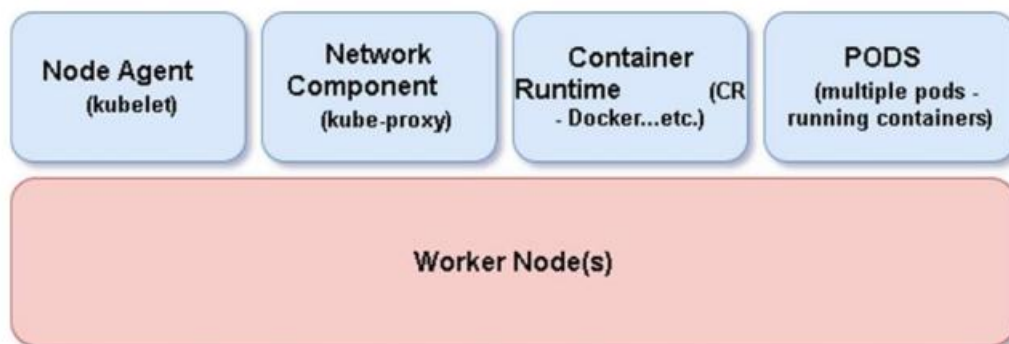
- etcd is the database of the cluster.
- **It stores:**
 - Pod definitions
 - Node information
 - Secrets
 - Configurations
- If etcd dies → cluster loses its memory.

3. Scheduler

- This is the decision maker.
- It decides:
- Which worker node should run this Pod?
- **It checks:**
 - CPU
 - Memory
 - Resources
- Then assigns the Pod to a node.

4. Controller Manager

- kubectl manager is responsible for overall health of the cluster.
- It make sures the actual state of kubernetes cluster matches to the desired state.
- User applications will not be scheduled unto master node. But they would work only on worker node.



1. Kubelet

- The job of kubelet is to make sure what is happening on the node and communicates it back to the master node.
- If a pod goes down, kubelet communicates to master node.

- This is the **agent** of Kubernetes on each worker.
- Talks to API Server, gets instructions, Starts containers, Reports status
- If kubelet stops, node becomes unhealthy.

2. Kube-proxy

- This handles networking.
- It:
 - Manages service IPs
 - Routes traffic
 - Helps pods talk to each other
- Without kube-proxy, networking breaks.

3. Pod

- A Pod is the smallest unit that runs on a Worker Node. A Pod contains one or more containers.

4. Container Runtime

- This is what actually runs containers.
- Example from your image:
- CRI-O
- It:
 - Pulls images
 - Creates containers
 - Runs them
- Kubernetes itself does NOT run containers.
The runtime does.

